

Data Viz Playground (Dashboard Graph Prep)

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Italy vs Poland VNL Grand Finals Heat Map

```
ball_data <- read_csv("C:/Users/RobTop/Documents/STA227 - Data Visualization/vb/ball_positions.csv")

## Rows: 267082 Columns: 7
## -- Column specification -----
## Delimiter: ","
## dbl (7): frame, u, v, xmin, ymin, xmax, ymax
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
ball_data$x_court <- -9 + ball_data$u * 18 # u in [0,1] mapped to [-9,9]
#ball_data$y_court <- ball_data$v * 9      # v in [0,1] mapped to [0,9]
y_min_clip <- 2.9
y_max_clip <- 9
ball_data$y_court <- pmax(pmin(ball_data$v * 9, y_max_clip), y_min_clip) # clip
ball_data$y_court <- (ball_data$y_court - y_min_clip) / (y_max_clip - y_min_clip) * 9 # stretch to 0-9

z <- kde2d(
  x = ball_data$x_court,
  y = ball_data$y_court,
  n = 200,                # higher resolution
  lims = c(-9, 9, 0, 9)
)
z$z <- sqrt(sqrt(z$z)) # more sensitive

# 3. Set up plot with extra right margin for legend
par(mar = c(2,2,2,6)) # bottom, left, top, right

# 4. Plot heatmap
image(
  z,
  col = hcl.colors(20, "YlOrRd", rev = TRUE), # more colors for sensitivity
  xlim = c(-9, 9),
  ylim = c(0, 9),
  axes = FALSE,
  xlab = "",
  ylab = "",
  main = "Volleyball Ball Heatmap"
)
```

```

# 5. Overlay court boundaries
rect(-9, 0, 9, 9, border = "black", lwd = 2) # outer boundary
abline(v = 0, lwd = 2) # centerline (net)
abline(v = -3, lty = 2) # attack lines
abline(v = 3, lty = 2)

# 6. Add half-court shading
rect(-9, 0, 0, 9, border = NA, col = rgb(0.9, 0.9, 1, 0.025))
rect(0, 0, 9, 9, border = NA, col = rgb(1, 0.9, 0.9, 0.025))

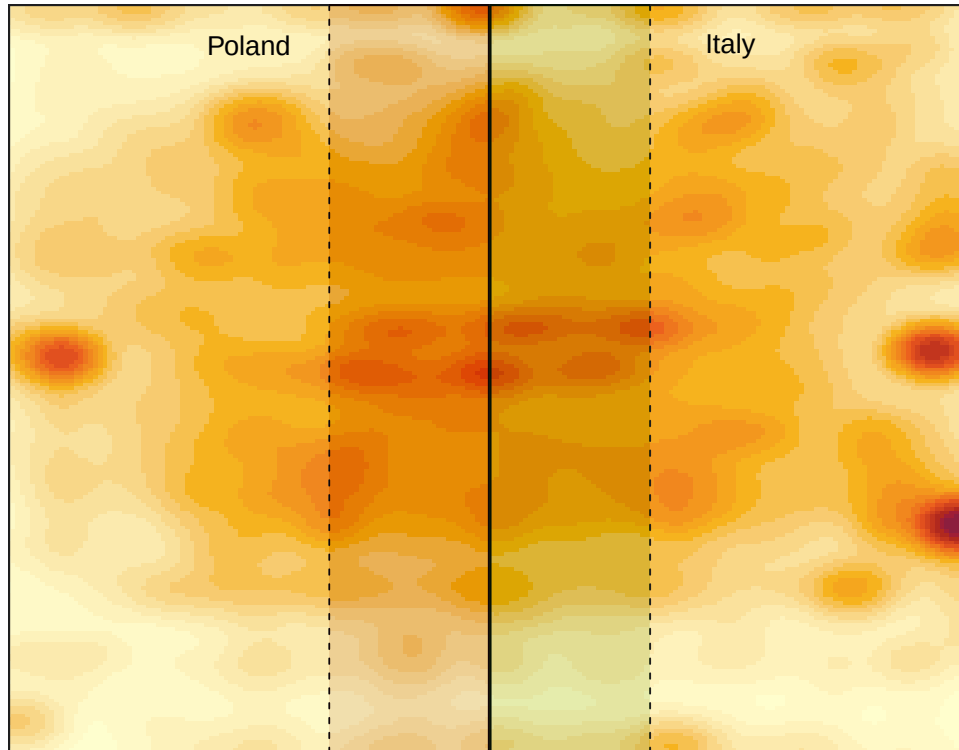
rect(-9, 0, -3, 9, border = NA, col = rgb(1, 1, 1, 0.1)) # Team A attack (dark red, transparent)
rect(-3, 0, 0, 9, border = NA, col = rgb(0.5, 0, 0, 0.1)) # Team A defense (white, transparent)
rect(3, 0, 9, 9, border = NA, col = rgb(1, 1, 1, 0.1)) # Team B attack (white, transparent)
rect(0, 0, 3, 9, border = NA, col = rgb(0, 0.5, 0, 0.1)) # Team B defense (dark green, transparent)

# 8. Optional team labels
text(-4.5, 8.5, "Poland", cex = 0.9)
text( 4.5, 8.5, "Italy", cex = 0.9)

# 9. Legend outside the court
cols <- hcl.colors(20, "YlOrRd", rev = TRUE)
legend_x <- 9.2
legend_y <- seq(0, 9, length.out = length(cols)+1)
for(i in 1:length(cols)){
  rect(legend_x, legend_y[i], legend_x + 0.5, legend_y[i+1], col = cols[i], border = NA)
}
# Add numerical labels for density
text(legend_x + 0.6, seq(0, 9, length.out=5),
     labels = round(seq(min(z$z), max(z$z), length.out=5), 2), adj = 0)

```

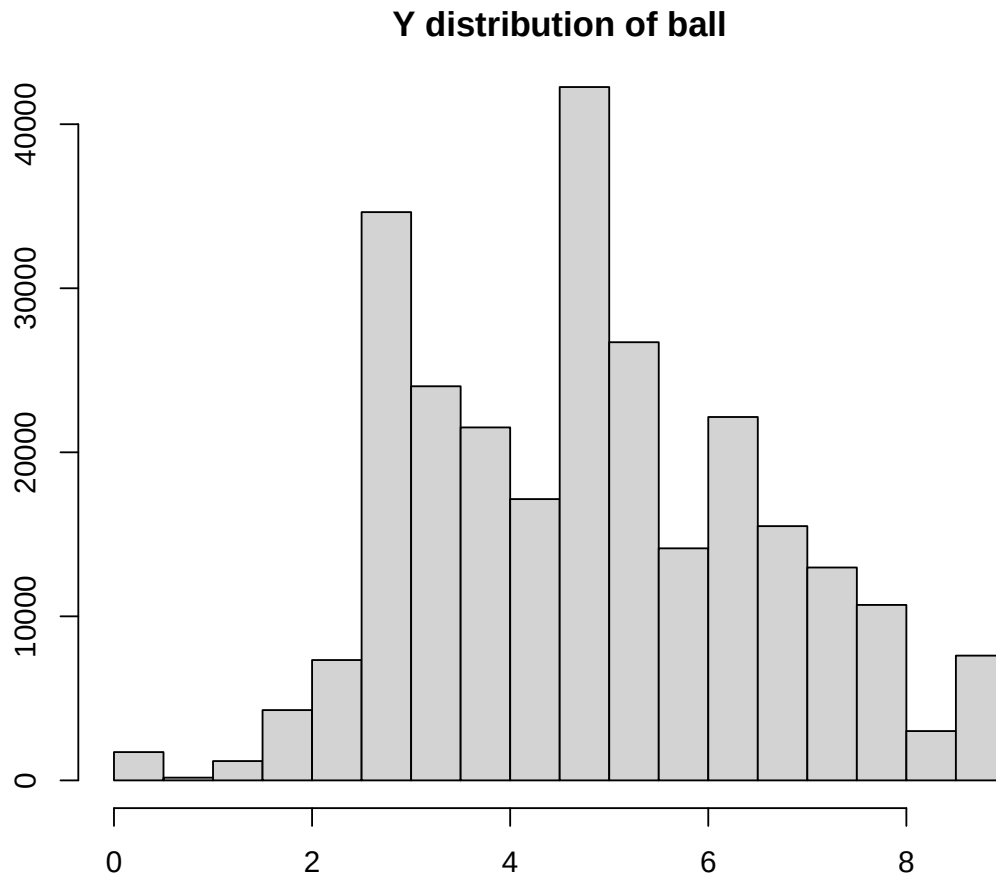
Volleyball Ball Heatmap



```
summary(ball_data$y_court)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000   3.357   4.737   4.816   6.107   9.000
```

```
hist(ball_data$y_court, breaks = 20, main="Y distribution of ball", xlab="Court Y")
```



Beach Volleyball - Beach VNL

Located on Kaggle: <https://www.kaggle.com/datasets/jessemostipak/beach-volleyball>

```
vb_data <- read_csv("C:/Users/RobTop/Documents/STA227 - Data Visualization/vb_matches.csv")
```

```
## Rows: 76756 Columns: 65
## -- Column specification -----
## Delimiter: ","
## chr  (17): circuit, tournament, country, gender, w_player1, w_p1_country, w...
## dbl  (42): year, match_num, w_p1_age, w_p1_hgt, w_p2_age, w_p2_hgt, l_p1_age...
## date  (5): date, w_p1_birthdate, w_p2_birthdate, l_p1_birthdate, l_p2_birthdate
## time  (1): duration
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
vb_clean <- vb_data %>% mutate(duration_min = as.numeric(hms(duration)) / 60, w_p1_age = as.numeric(w_p1...
head(vb_data)
```

```
## # A tibble: 6 x 65
##   circuit tournament country year date gender match_num w_player1
##   <chr> <chr> <chr> <dbl> <date> <chr> <dbl> <chr>
## 1 AVP Huntington Beach United S~ 2002 2002-05-24 M 1 Kevin Wo~
## 2 AVP Huntington Beach United S~ 2002 2002-05-24 M 2 Brad Tor~
## 3 AVP Huntington Beach United S~ 2002 2002-05-24 M 3 Eduardo ~
```

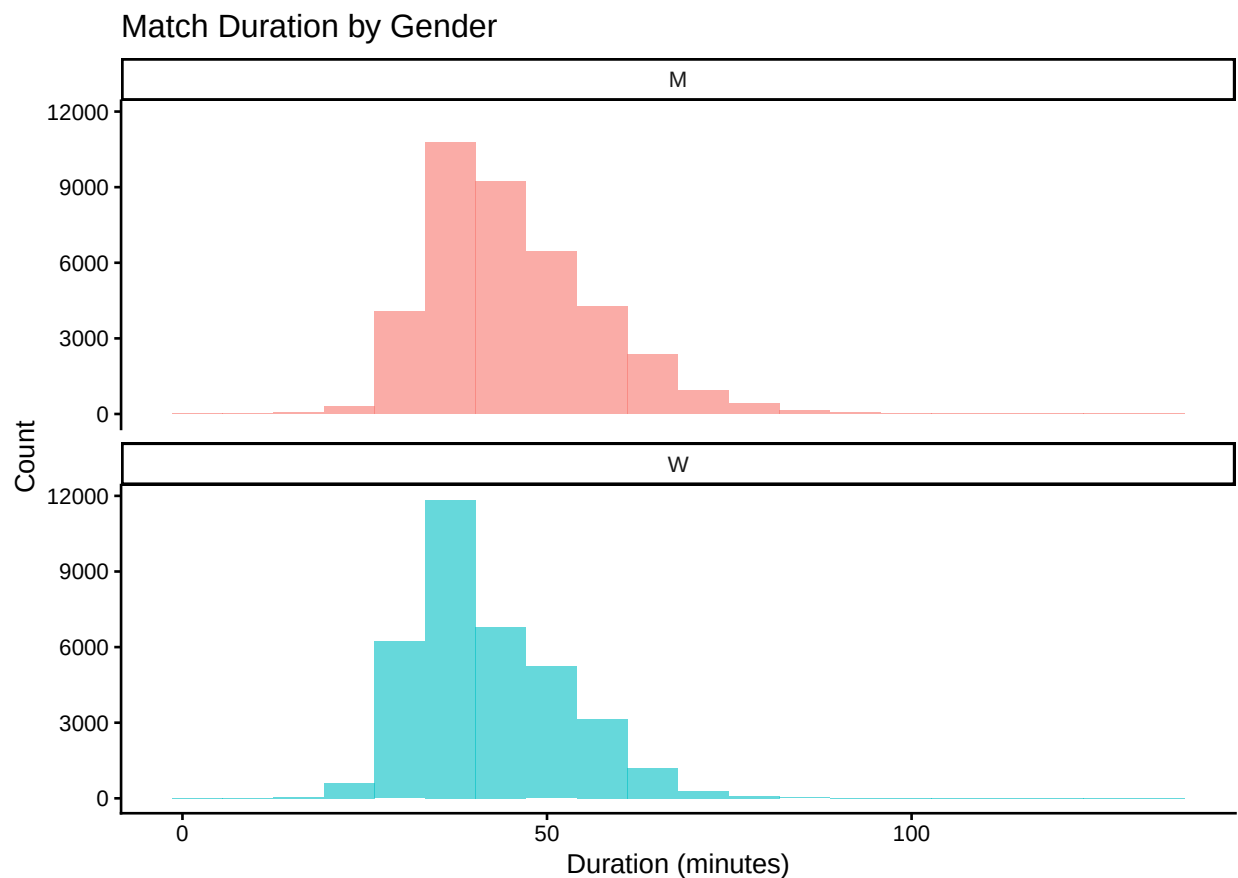
```
## 4 AVP      Huntington Beach United S~ 2002 2002-05-24 M          4 Brent Do~
## 5 AVP      Huntington Beach United S~ 2002 2002-05-24 M          5 Albert H~
## 6 AVP      Huntington Beach United S~ 2002 2002-05-24 M          6 Jason Ri~
## # i 57 more variables: w_p1_birthdate <date>, w_p1_age <dbl>, w_p1_hgt <dbl>,
## #   w_p1_country <chr>, w_player2 <chr>, w_p2_birthdate <date>, w_p2_age <dbl>,
## #   w_p2_hgt <dbl>, w_p2_country <chr>, w_rank <chr>, l_player1 <chr>,
## #   l_p1_birthdate <date>, l_p1_age <dbl>, l_p1_hgt <dbl>, l_p1_country <chr>,
## #   l_player2 <chr>, l_p2_birthdate <date>, l_p2_age <dbl>, l_p2_hgt <dbl>,
## #   l_p2_country <chr>, l_rank <chr>, score <chr>, duration <time>,
## #   bracket <chr>, round <chr>, w_p1_tot_attacks <dbl>, ...
```

```
# Graph 1: Match duration by gender
```

```
gf_histogram(~ duration_min, fill = ~ gender, data = vb_clean, bins = 20, alpha = 0.6) %>% gf_facet_wrap
```

```
## Warning: Removed 2249 rows containing non-finite outside the scale range
```

```
## (`stat_bin()`).
```



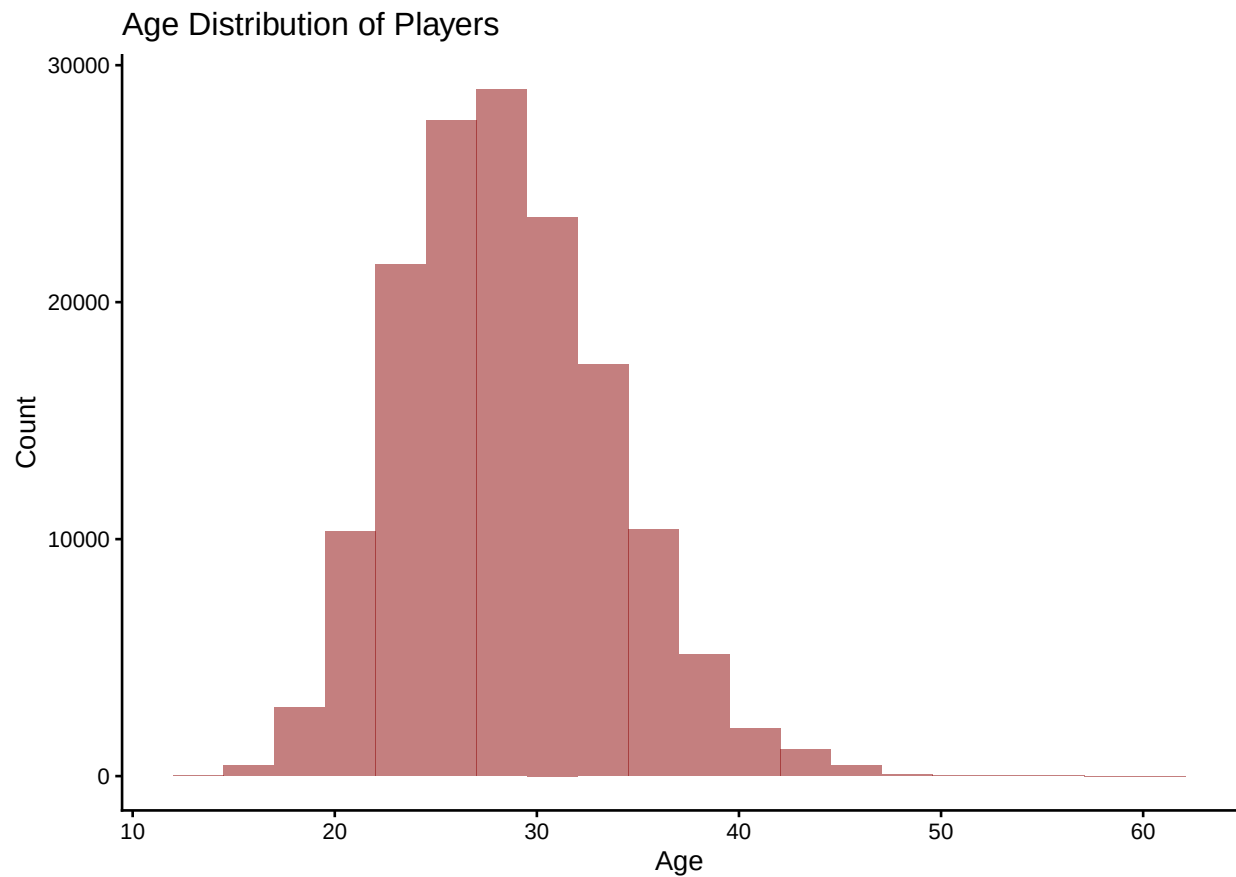
```
# Graph 2: Age distribution
```

```
age_data <- data.frame(age = c(vb_clean$w_p1_age, vb_clean$l_p1_age))
```

```
gf_histogram(~ age, data = age_data, bins = 20, fill="darkred") %>% gf_labs(title = "Age Distribution of
```

```
## Warning: Removed 1442 rows containing non-finite outside the scale range
```

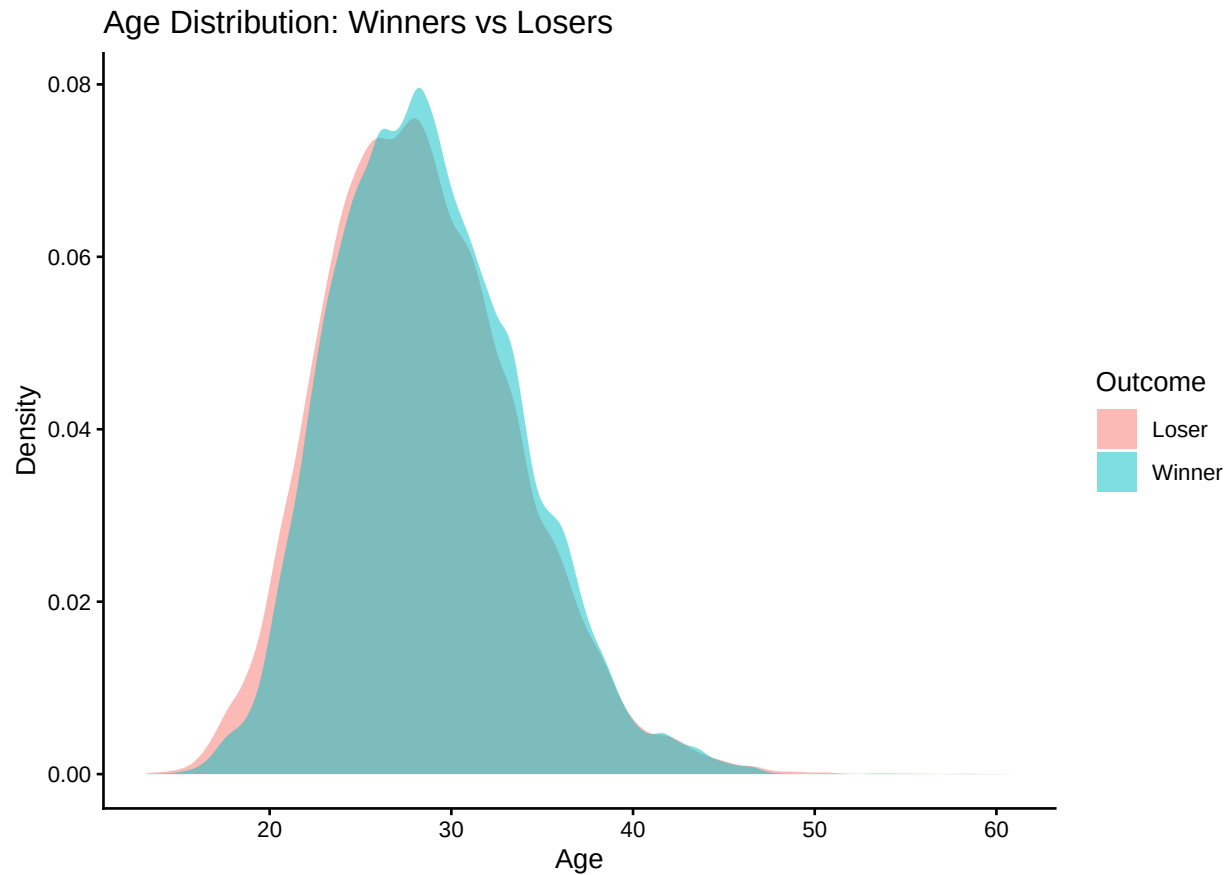
```
## (`stat_bin()`).
```



Graph 3: Age comparison winners vs losers

```
all_ages <- data.frame(age = c(vb_clean$w_p1_age, vb_clean$l_p1_age), outcome = rep(c("Winner", "Loser"))
gf_density(~ age, fill = ~ outcome, data = all_ages, alpha=.5) %>% gf_labs(title = "Age Distribution: Win
```

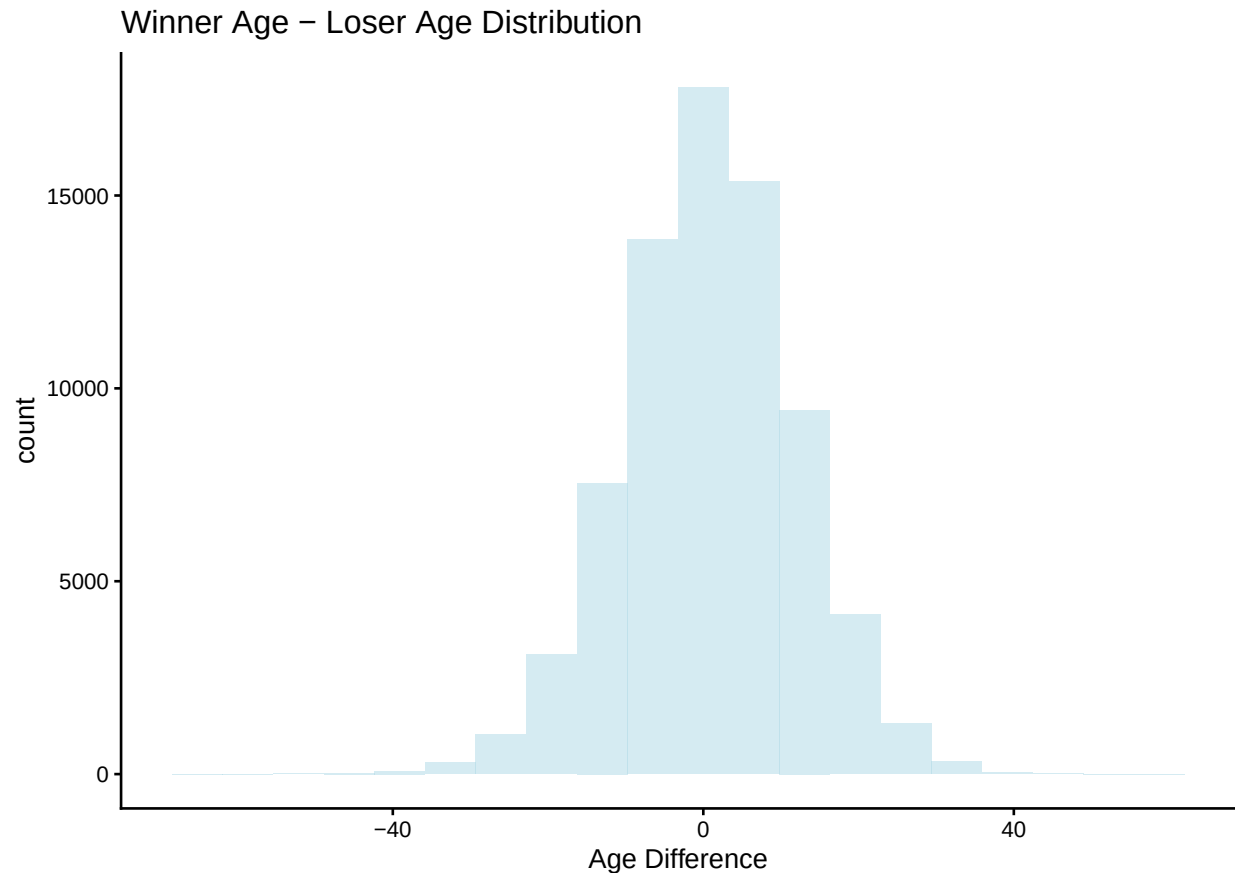
```
## Warning: Removed 1442 rows containing non-finite outside the scale range
## (`stat_density()`).
```



Graph 4: Age Difference

```
vb_clean = vb_clean %>%mutate(age_diff = w_p1_age + w_p2_age - l_p1_age - l_p2_age)
gf_histogram(~age_diff, data = vb_clean, bins = 20, fill = "lightblue") %>% gf_labs(title = "Winner Age

## Warning: Removed 2347 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



PlusLiga '08-'23 - Indoor Volleyball

Data on Kaggle: <https://www.kaggle.com/datasets/kacpergregorowicz/-mens-volleyball-plusliga-20082022>

```
plugliga <- read_csv("C:/Users/RobTop/Documents/STA227 - Data Visualization/Mens-Volleyball-PlusLiga-2008-2023.csv")
```

```
## Rows: 2639 Columns: 44
## -- Column specification -----
## Delimiter: ","
## chr (19): Date, Team_1, Team_2, T1_Srv_Err, T1_Srv_Eff, T1_Rec_Pos, T1_Rec_P...
## dbl (25): T1_Score, T2_Score, T1_Sum, T1_BP, T1_Ratio, T1_Srv_Sum, T1_Srv_Ac...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
plugliga <- plugliga %>% mutate(Date = dmy_hm(Date), Year = year(Date), Winner = factor(Winner, levels = c("L", "W")))
```

```
## Warning: There were 2 warnings in `mutate()`.
## The first warning was:
## i In argument: `T1_Att_Eff = as.numeric(T1_Att_Eff)`.
## Caused by warning:
## ! NAs introduced by coercion
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```
head(plugliga)
```

```
## # A tibble: 6 x 45
##   Date           Team_1      Team_2 T1_Score T2_Score T1_Sum T1_BP T1_Ratio
##   <dtm>          <chr>      <chr>    <dbl>    <dbl>    <dbl> <dbl>    <dbl>
## 1 2022-10-01 14:45:00 AZS Olsztyn ZAKSA~      1         3     60     17      11
## 2 2022-09-30 17:30:00 Jastrzębsk~ GKS K~      3         0     51     17      27
## 3 2022-10-01 20:30:00 LUK Lublin Czarn~      2         3     76     23      35
## 4 2022-10-02 14:45:00 Warta Zawie~ PGE S~      3         2     66     16      22
## 5 2022-10-03 17:30:00 BBTS Bielsk~ Cupru~      1         3     63     22      17
## 6 2022-10-02 20:30:00 Stal Nysa Śleps~      3         1     68     23      29
## # i 37 more variables: T1_Srv_Sum <dbl>, T1_Srv_Err <chr>, T1_Srv_Ace <dbl>,
## #   T1_Srv_Eff <chr>, T1_Rec_Sum <dbl>, T1_Rec_Err <dbl>, T1_Rec_Pos <chr>,
## #   T1_Rec_Perf <chr>, T1_Att_Sum <chr>, T1_Att_Err <dbl>, T1_Att_Blz <dbl>,
## #   T1_Att_Kill <dbl>, T1_Att_Kill_Perc <chr>, T1_Att_Eff <dbl>,
## #   T1_Blz_Sum <dbl>, T1_Blz_As <chr>, T2_Sum <dbl>, T2_BP <dbl>,
## #   T2_Ratio <dbl>, T2_Srv_Sum <dbl>, T2_Srv_Err <chr>, T2_Srv_Ace <dbl>,
## #   T2_Srv_Eff <chr>, T2_Rec_Sum <dbl>, T2_Rec_Err <dbl>, T2_Rec_Pos <chr>, ...
```

```
# Graph 1: Serve effectiveness comparison
```

```
gf_point(jitter(as.numeric(T2_Srv_Eff)) ~ jitter(as.numeric(T1_Srv_Eff)), data = plugliga, color = ~Win
```

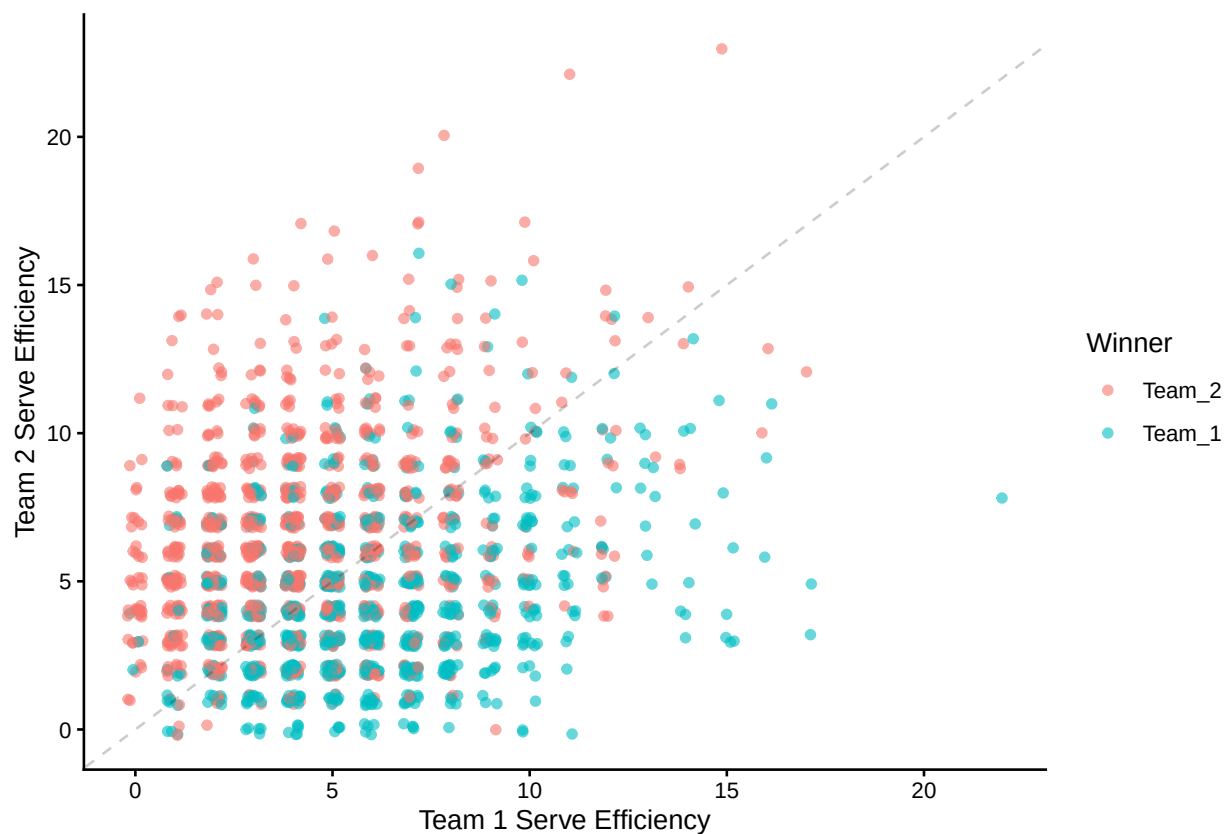
```
## Warning in jitter(as.numeric(T1_Srv_Eff)): NAs introduced by coercion
```

```
## Warning in jitter(as.numeric(T2_Srv_Eff)): NAs introduced by coercion
```

```
## Warning: Removed 628 rows containing missing values or values outside the scale range
```

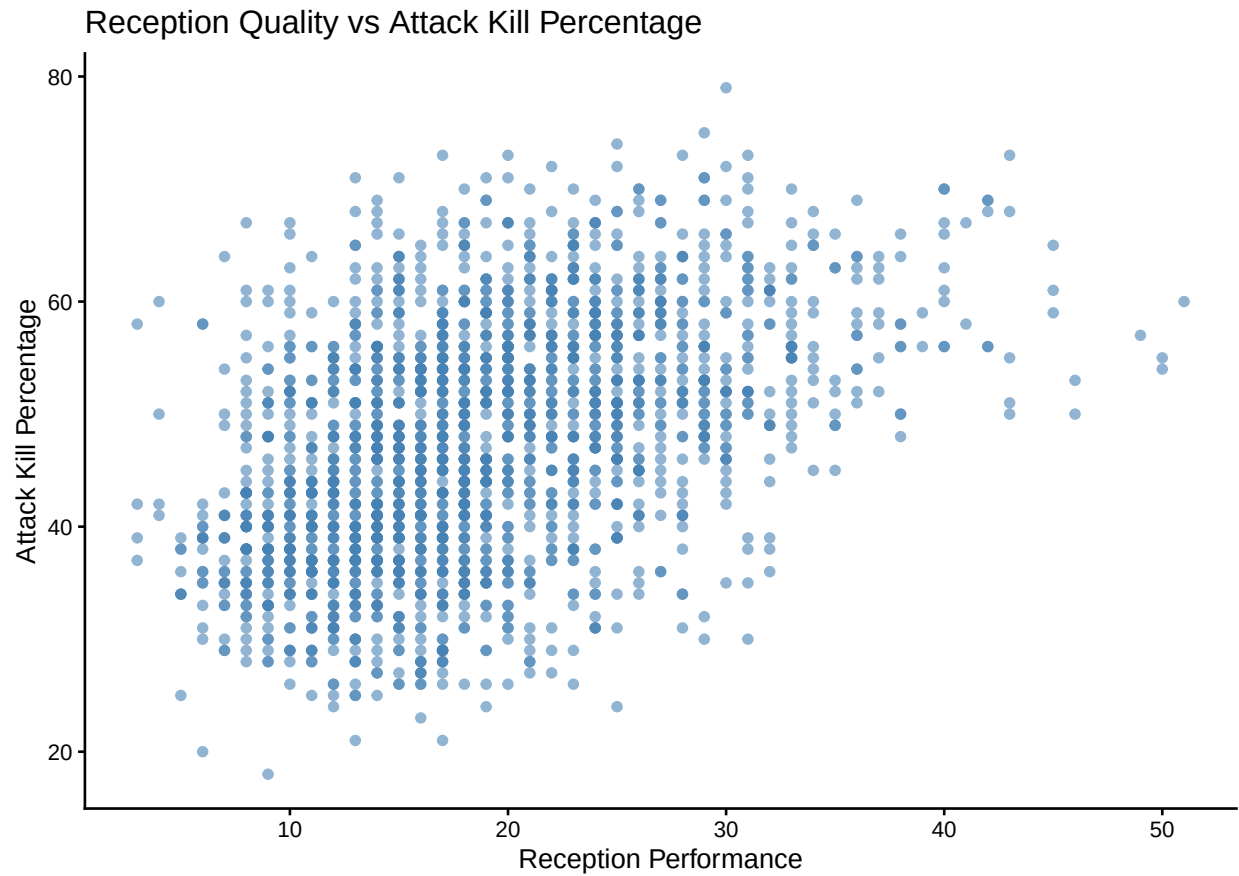
```
## (`geom_point()`).
```

Serve Effectiveness: Team 1 vs Team 2



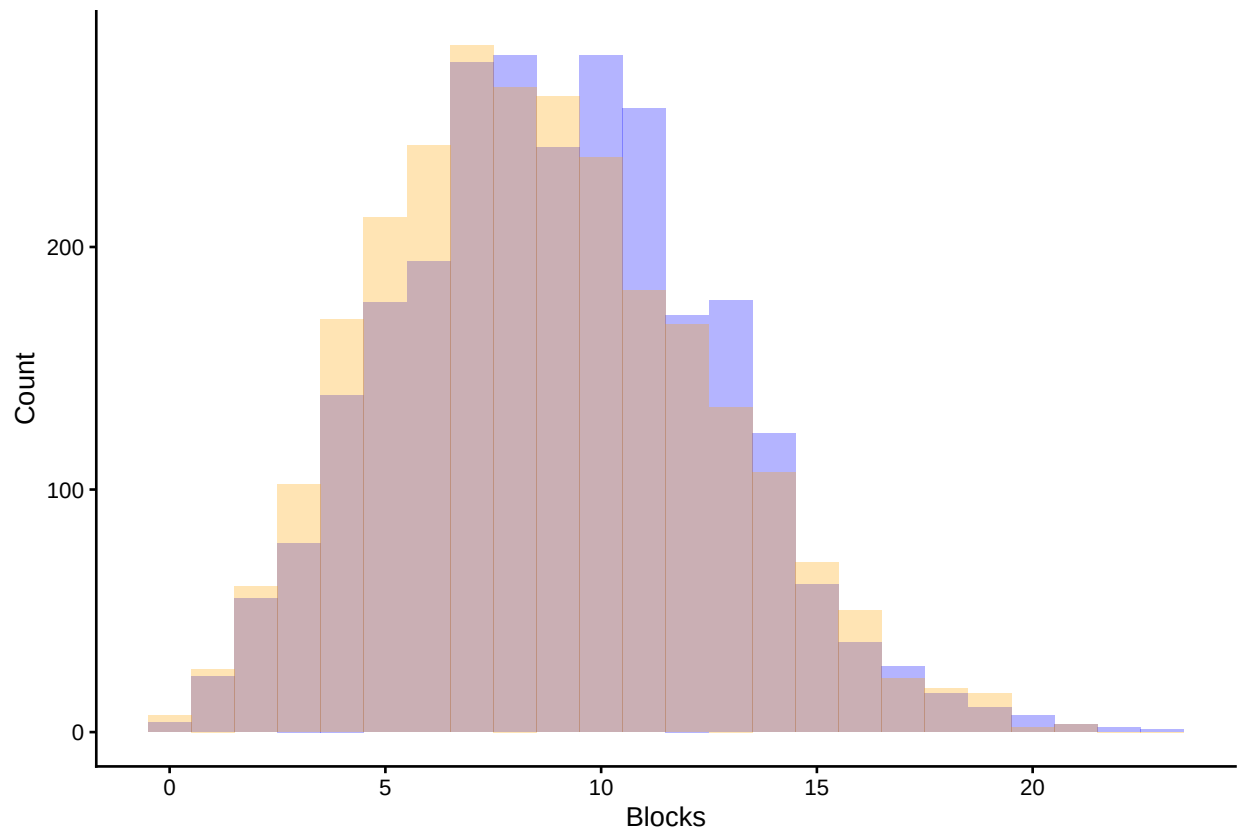
```
# Graph 2: Reception vs attack
gf_point(as.numeric(T1_Att_Kill_Perc) ~ as.numeric(T1_Rec_Perf), data = plugliga, alpha = 0.6, color = "blue")

## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning: Removed 628 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
# Graph 3: Blocking performance Overlay
gf_histogram(~T1_Blck_Sum, data = plugliga, fill = "blue", alpha = 0.3, binwidth = 1) %>% gf_histogram(~T1_Rec_Perf, data = plugliga, fill = "red", alpha = 0.3, binwidth = 1)
```

Distribution of Blocks – Both Teams Overlaid



```
# Graph 4: Blocking performance Summed
gf_histogram(~(T1_Blkc_Sum + T2_Blkc_Sum), data = plugliga, alpha = 0.5, position = "stack", binwidth = 1,
```

